

Grid Ways Problem Description

You are standing at the:

top-left corner

of a grid.

Your goal is to reach the:

bottom-right corner

You can move only:

- Right
- Down

The task is:

Find total number of possible ways to reach destination.

Example

For a 3×3 grid:

(0,0) (0,1) (0,2)

(1,0) (1,1) (1,2)

(2,0) (2,1) (2,2)

Start:

$(0, 0)$

Destination:

$(2, 2)$

Allowed moves:

→ Right
↓ Down

Possible Paths

RRDD
RDRD
RDDR
DRRD
DRDR
DDRR

Total:

6 ways

Core Recursive Thinking

At every cell:

You have only 2 choices:

1. Move Down
2. Move Right

So:

```
Total Ways =  
Ways from Down  
+  
Ways from Right
```

That becomes:

```
grid(i,j)  
=  
grid(i+1,j)  
+  
grid(i,j+1)
```

Meaning of Function

```
grid(i,j,n,m)
```

means:

```
Number of ways to reach destination  
starting from cell (i,j)
```

Base Cases

1. Last Row OR Last Column

```
if(i==n-1 || j==m-1)
    return 1;
```

Reason:

If you are already in:

- last row → only move RIGHT possible
- last column → only move DOWN possible

So only:

```
1 path remains
```

2. Outside Grid

```
if(i==n || j==m)
    return 0;
```

Reason:

Outside grid means:

```
invalid path
```

Real-Life Analogy

Imagine:

- You are in a city grid.
- You can only move:
 - east
 - south

Question:

How many different routes exist to reach destination?

Important Concepts Learned

This problem teaches:

- Recursion
- Divide and Conquer
- Overlapping Subproblems
- Dynamic Programming foundation
- Recursion Tree
- Path Counting

Important Observation

Many states repeat.

Example:

`grid(1,1)`

gets calculated multiple times.

This redundancy is why:

- recursion becomes slow
- DP optimization is needed

Time Complexity

Pure recursion:

$$O(2^{(n+m)})$$

because each call splits into 2 more calls.

Optimized Approaches

1. Memoization

Store already solved states.

2. Tabulation

Build answer bottom-up using DP table.

3. Mathematical Formula

Total moves needed:

(n-1) downs
(m-1) rights

Total moves:

$$(n+m-2)$$

Ways:

$$(n+m-2)! / ((n-1)! * (m-1)!)$$

Combination formula.

One-Line Summary

Grid Ways problem finds the total number of paths from top-left to bottom-right.